

AI / DEVOPS / SUSTAINABILITY

Supply Chain Optimizer Team : TeamCMPICA

Predictive Logistics & Real-Time Cold Chain Mitigation Platform

PART 1

Supply Chain Fragility

Cascading disruptions, unmonitored temperature breaches, and idle fleet assets across global transit networks.

The Costly Reality of Logistics

Who & What is Affected

- ⚠️ **Cascading Hazard Events:** Weather anomalies, port congestion, and strikes impact hundreds of active shipments simultaneously.
- 👥 **Logistics & Fleet Operators:** Suffer from zero real-time visibility, leading to unallocated idle trucks while key corridors bottleneck.
- 📡 **Cold Chain Vulnerability:** Thermal excursions across intermediate transit legs go undetected until final delivery.



Impact of Unmonitored Spoilage

\$500K+

Loss Per Spoiled Cargo

Why Legacy Systems Fail

- 🕒 **Passive Logging:** Point-in-time checks discover thermal breaches after goods are already destroyed.
- 🚫 **No Proactive Mitigation:** Legacy monitoring tools provide zero automated re-routing or hazard warning capabilities.
- 🌿 **Sustainability Waste:** Idle fleet assets waste fuel while spoiled perishables generate massive landfill overhead.

PART 2

Predictive Supply Chain AI



A unified platform for continuous surveillance, dynamic re-routing, and automated asset rebalancing.

How Supply Chain Optimizer Works



1. Real-Time Telemetry Ingestion

Telemetry from containers and vehicles streams through modular FastAPI endpoints (`shipments.py`, `cold_chain.py`) for immediate validation and ORM storage.



2. Dynamic Risk & Fleet Mitigation

The disruption engine (`disruptions.py`) evaluates environmental hazards against active routes, auto-rerouting shipments and rebalancing idle assets (`fleet.py`).

Thermal & Fleet Safeguards

Continuous Thermal Surveillance

Active sensors track internal container temperatures against strict compliance thresholds, triggering immediate re-routing long before cargo damage occurs.

Fleet Asset Rebalancing

Idle trucks and vessels are automatically matched to overloaded corridors, reducing empty miles and fuel consumption.



High-Performance Tech Stack



FastAPI Backend

Asynchronous modular architecture with dedicated Python REST routes for high-throughput telemetry processing.



Structured SQL Layer

Robust relational schema design (`schema.sql``) enforcing data integrity and rapid spatial queries across shipments.



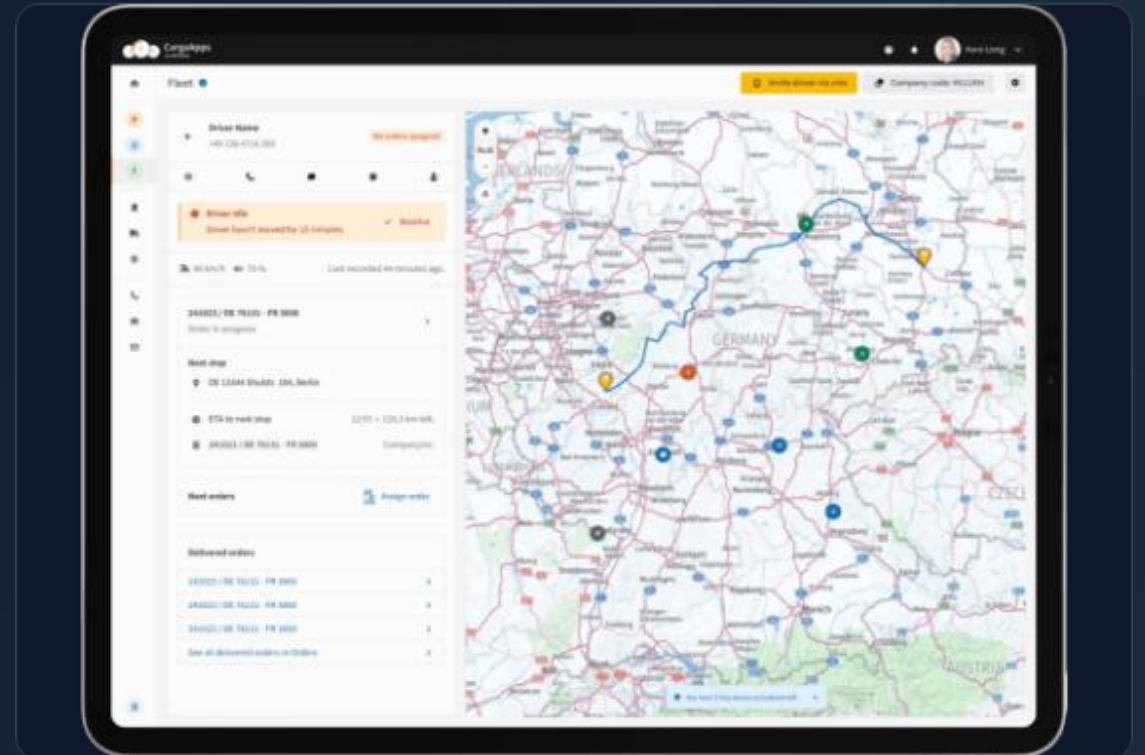
Interactive Dashboard

Single-pane visual control dashboard (`index.html``, `app.js``) displaying live risk scores and fleet telemetry.

IBM watsonx & Cloud Integration

Powered by IBM AI & Cloud

-  **IBM watsonx.ai:** Evaluates historical weather and disruption data to predict route risks and thermal excursion probabilities.
-  **IBM Cloud Code Engine:** Hosts serverless backend containers for seamless horizontal scaling under traffic spikes.
-  **IBM Secrets Manager:** Securely handles API keys and database credentials across environments.



Core Component Breakdown

Layer	Technology	Core Responsibility
Frontend UI	JavaScript / HTML5 / CSS3	Renders real-time shipment maps, asset status, and alert popups.
API Engine	Python FastAPI / Pydantic	Validates telemetry payloads and routes request pipelines.
AI Model Layer	IBM watsonx.ai SDK	Predicts disruption severities and optimal alternate transit paths.
Persistence	PostgreSQL / SQLite	Stores relational records for shipments, cold chain logs, and fleet assets.

Beyond the Hackathon



Decarbonization

Eliminates empty-mile trips and cargo spoilage, directly reducing supply chain carbon emissions.



Carrier IoT Mesh

Seamless integration with global ocean line carriers and satellite IoT temperature sensors.



Autonomous Dispatch

Automated smart contracts that instantly trigger rerouting and insurance claims on breach detection.